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Rares-Darius Buhai

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Current Position

2025–present Postdoc, School of Computer and Communication Sciences, EPFL.
Host: Prof. Ola Svensson (ola.svensson@epfl.ch).

Education

2020–2025 **Dr.Sc.**, Department of Computer Science, ETH Zurich.
Advisor: Prof. David Steurer (dsteurer@ethz.ch).
Thesis: *Efficient Estimation and Certification in High-Dimensional Semi-random and Non-Spherical Models*.

2019–2020 **M.Eng.**, Department of Electrical Engineering and Computer Science, MIT.
Advisor: Prof. Guy Bresler (guy@mit.edu).

2015–2019 **B.Sc.**, Department of Electrical Engineering and Computer Science, MIT.

Industry Experience

2017 Software Engineering Intern, Google.

2016 Software Engineering Intern, Facebook.

Awards

Gold medal at the International Olympiad in Informatics (IOI), years 2012, 2013, 2014, and 2015.

Teaching

Teaching assistant for:

- ETH Zurich: Algorithms and Data Structures (Fall 2021, 2022, 2023, 2024), Algorithmic Foundations of Data Science (Spring 2022), Optimization for Data Science (Spring 2021).
- MIT: Introduction to Probability 6.041 (Spring 2020), Machine Learning 6.867 (Fall 2019).

Academic Service

Reviewed for STOC, FOCS, SODA, COLT, ITCS, NeurIPS, APPROX, ESA, and TCC.

Publications

- **Optimal Algorithms and Hardness for Max-Cut in Bounded Degree Graphs**. Rares-Darius Buhai, Suprovat Ghoshal, Neng Huang, Pravesh K. Kothari, Euiwoong Lee, Konstantin Makarychev, Yury Makarychev. SODA 2027.

- **Dimension Reduction via Sum-of-Squares and Improved Clustering Algorithms for Non-Spherical Mixtures**.
  Prashanti Anderson, Mitali Bafna, Rares-Darius Buhai, Pravesh K. Kothari, David Steurer. COLT 2026.

- **Finding Colorings in One-Sided Expanders**.
  Rares-Darius Buhai, Yiding Hua, David Steurer, Andor Vári-Kakas. FOCS 2025.

- **The Quasi-Polynomial Low-Degree Conjecture Is False**.
  Rares-Darius Buhai, Jun-Ting Hsieh, Aayush Jain, Pravesh K. Kothari. FOCS 2025.

- **Robust Mixture Learning when Outliers Overwhelm Small Groups**.
  Daniil Dmitriev, Rares-Darius Buhai, Stefan Tiegel, Alexander Wolters, Gleb Novikov, Amartya Sanyal, David Steurer, Fanny Yang. NeurIPS 2024.

- **Semirandom Planted Clique and the Restricted Isometry Property**.
  Jarosław Błasiok, Rares-Darius Buhai, Pravesh K. Kothari, David Steurer. FOCS 2024.

- **Computational-Statistical Gaps for Improper Learning in Sparse Linear Regression**.
  Rares-Darius Buhai, Jingqiu Ding, Stefan Tiegel. COLT 2024.

- **Beyond Parallel Pancakes: Quasi-Polynomial Time Guarantees for Non-Spherical Gaussian Mixtures**.
  Rares-Darius Buhai, David Steurer. COLT 2023.

- **Algorithms Approaching the Threshold for Semi-Random Planted Clique**.
  Rares-Darius Buhai, Pravesh K. Kothari, David Steurer. STOC 2023.

- **Learning Restricted Boltzmann Machines with Sparse Latent Variables**.
  Guy Bresler, Rares-Darius Buhai. NeurIPS 2020.

- **Empirical Study of the Benefits of Overparameterization in Learning Latent Variable Models**.
  Rares-Darius Buhai, Yoni Halpern, Yoon Kim, Andrej Risteski, David Sontag. ICML 2020.

- **Human-Aware Robotic Assistant for Collaborative Assembly: Integrating Human Motion Prediction With Planning in Time**.
  Vaibhav V. Unhelkar, Przemyslaw A. Lasota, Quirin Tyroller, Rares-Darius Buhai, Laurie Marceau, Barbara Deml, Julie A. Shah. IEEE Robotics and Automation Letters 2018.

Preprints

- **A Better-than-2 Approximation for k-Median Part II: Stable Instances**.
  Rares-Darius Buhai, Vincent Cohen-Addad, Henry Fleischmann, Anupam Gupta, Antoni Jubés Monforte, Madhusudhan Reddy Pittu, Miltiadis Stouras, Ola Svensson, Lukas Vogl. Preprint 2026.

- **A Better-than-2 Approximation for k-Median Part I: The Power of General Bids**.
  Rares-Darius Buhai, Vincent Cohen-Addad, Henry Fleischmann, Anupam Gupta, Antoni Jubés Monforte, Madhusudhan Reddy Pittu, Miltiadis Stouras, Ola Svensson, Lukas Vogl. Preprint 2026.

- **Efficient Recovery of Latent Coordinate Structure from Sparse Observations of the Hypercube**.
  Rares-Darius Buhai, Davide Mazzali, Weronika Wrzos-Kaminska. Preprint 2026.
- **Lasso and Partially-Rotated Designs**.
  Rares-Darius Buhai. Preprint 2025.